TECHNICAL EVALUATION REPORT

Koopman Spectral Theory for Indivisible Stochastic Processes

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1 Summary of the submission

The paper develops spectral machinery for two-time transition families $\{P(t,s)\}_{s\le t}$ on $L^2(\Omega,\mu)$ which violate the Chapman–Kolmogorov (CK) condition irreducibly, and for which no time-ordered-exponential generator in the Dyson (Bochner-integrable) class exists. The carrier of dynamical information is the composition deficit

$$\Delta(t,s,r) \;=\; P(t,s) - P(t,r)P(r,s), \qquad 0 \le s < r < t \le T, \qquad D(t,s,r) := \|\Delta(t,s,r)\|_{\mathrm{op}}, \quad (1)$$

and the framework is built throughout on the *singular* structure of the deficit family rather than on eigenexpansions. The author enumerates seven contributions in Section 1 of the paper. (Throughout this report the section sign refers to the manuscript under review, never to this report.) In my assessment they are of markedly unequal weight; the paper itself says so for two of them, and the assessment of originality below sets out where the weight actually lies.

The classical setting rests on a composition law. For a stochastic transition family that law is CK factorization, and the manuscript asks what operator-theoretic structure survives when it fails at the two-time level. Rather than forcing the data into a generator-based model, it promotes the failure of composition itself to the central object. This is an intellectually coherent choice: if the missing cocycle property is exactly the hypothesis that licenses generator-based spectral machinery, then the defect of that property is a natural candidate for carrying the missing dynamical information. The paper is therefore to be judged by whether it extracts mathematically nontrivial and useful structure from Δ rather than merely renaming CK failure.

The substantive strands are: global input and output deficit subspaces with a universal three-pair obstruction bound; a gauge-relative decomposition of the deficit into an interference cross-term and a squared-modulus gap, with the counterexamples that prevent either term being read in isolation; covariant eigenfamily diagnostics; a local second-order *indivisibility tensor* $\beta(s)$ with an exact triple-scaling law; a Stinespring unitary lift with a finite-grid cocycle realization for divisible families; and an integrated trajectory-space deficit $J(R)$. A causal reformulation on $L^2([0,T]\times\Omega)$ (§7) and a sketch of an empirical protocol (§8) complete the paper.

2 Overall assessment

This is competent and carefully bounded mathematical work with a genuine new invariant at its centre. The technical findings supporting that assessment, and the qualifications that attach to it, are set out below.

The strongest original element is not the use of any one standard technique (singular-value decomposition, Stinespring dilation, Koopman operators, or norm inequalities) but the organization of these tools into a coherent defect geometry for non-cocyclic stochastic transition families. The local tensor β with its exact second-order scaling law, together with the global deficit subspace geometry, carry the greatest weight.

The indivisibility tensor and its scaling law (Proposition 4.12) are correct in the general non-commuting, genuinely non-autonomous case. This was verified independently in this report rather than relying on the author's argument alone; the verification record is set out below. The gauge-relative decomposition (Theorem 4.8) is accompanied by exactly the two counterexamples needed to prevent it from being over-read.

What I regard as the strongest indicator of quality is not any single theorem but the discipline of the hypotheses. The standing contraction assumption is stated explicitly rather than assumed tacitly, and the paper later exhibits a family within its own example set for which it fails. The

non-existence result is scoped to the Dyson class. The finite-grid Stinespring cocycle construction is stated at the level actually proved, with the continuum problem left open. The unistochasticity discussion uses a valid dimension-dependent counterexample. The continuity hypothesis of Theorem 7.4 is imposed on the open region $\{s < t\}$, for a reason the paper states and have been also checked in this report (Finding 3): the closed-simplex version would be unsatisfiable in infinite dimensions under this framework's own assumptions, and the theorem would be vacuous in precisely the setting it targets.

Against this, three limitations should be recorded, none of them a defect of correctness.

First, and most consequential for revision: the priority claim must be stated carefully. Several mathematical ingredients and neighbouring constructions have clear precedents, and after a dedicated external priority audit the most defensible originality claim is a novel synthesis plus specific new diagnostics, rather than the invention of the underlying operator-theoretic primitives. The basic embedding of a classical stochastic matrix into a CPTP map, in particular, should not be claimed at all: it is anticipated in work predating this paper's deposit (Finding 5 below).

Second, part of the paper's declared contribution consists of short consequences of standard facts presented with disproportionate weight: Proposition 3.6 follows in two lines from the multiplicativity of time-ordered products, Theorem 4.5(a) is immediate from $\ker \Delta = \mathrm{Range}(\Delta^*)^{\perp}$, and Corollary 4.6 is a triangle inequality. The paper already applies the correct remedy elsewhere — Proposition 2.6 and Corollary 2.7 are explicitly relabelled as scope-fixing background — and the same treatment should be extended. Relatedly, the central dilation theorem of §5 is substantially an assembly of standard material around one new indexing device, and its presentation does not distinguish the new ingredient from the imported machinery as sharply as the paper's own Remark 5.6 distinguishes levels of rigour.

Third, the framework has not been exercised on any example where the answer is not evident in advance: the demonstrations are two- and three-state, and the estimator that would operationalise the empirical protocol is listed as open.

I found no mathematical error in any claim I verified, and no material defect. I found one attribution gap and two uncited prior-art items bearing on §5, all recorded under priority below. This report also contains a proposed resolution in the author's favour a question the paper leaves open.

3 Section-by-section analysis

3.1 Framework and scope (§2)

The definitions of transition kernel, CK condition, divisibility and deficit are standard and correctly given. Remark 2.5 is a useful piece of hygiene: results are stated for *transition families* and not for processes, and the question whether a given family is realizable by a genuine process is identified as a distinct structural problem and placed outside the paper's scope.

Proposition 2.6 and Corollary 2.7 fix the hierarchy — Markov $\Rightarrow$ CK, the converse failing by Feller's 1959 example — and are labelled classical, "presented as scope-fixing background rather than claimed as a contribution." The distinction drawn is correct and important: the Markov property constrains the full hierarchy of finite-dimensional distributions, whereas CK constrains only the two-time conditional kernels. The paper also acknowledges that unrestricted history augmentation can Markovize a process, while treating constrained augmentation as a separate modelling question. The self-labelling is accurate and is good practice.

Proposition 2.11 (unistochasticity). Every unistochastic matrix is doubly stochastic; at $n = 2$ the converse holds; for $n \geq 3$ both inclusions

$$\{\text{unistochastic}\} \subsetneq \{\text{doubly stochastic}\} \subsetneq \{\text{column-stochastic}\}$$

are strict. I have verified this. The $n = 3$ witness $\frac{1}{2}(J_3 - I_3)$ is standard, and the orthogonality argument — the first two rows would require a vanishing sum of three complex numbers of moduli $0, 0, \frac{1}{2}$ — is correct. The extension to $n > 3$ is handled by the padded witness $\frac{1}{2}(J_3 - I_3) \oplus I_{n-3}$ with a block-diagonalisation proof, and the author correctly notes in passing that the unpadded $\frac{1}{2}(J_n - I_n)$ has row sums $(n-1)/2$ and is therefore doubly stochastic only at $n = 3$. Both are correct, and the counterexample is properly dimension-dependent.

Remark 2.12. The identity $R(90°) = R(45°)R(45°)$ exhibits an exact unitary cocycle whose squared moduli satisfy $F^2 = F \neq X$: a perfect cocycle with maximal CK violation. Correct, and an effective demonstration that unitary cocycle structure and CK divisibility are logically independent in both directions.

3.2 Conventions and the breakdown of semigroup structure (§3)

Assumption 3.3 (standing contraction) is stated explicitly, with Remark 3.4 supplying Schur-test sufficient conditions; §6.7 later exhibits a family within the paper's own example set for which the assumption fails, which I regard as a point in the author's favour rather than against him. Remark 3.5 records the Hilbert-bundle formulation as the natural general setting.

Proposition 3.6 (non-existence of a generator) is correct, and is scoped explicitly to the class actually established. A time-ordered exponential generated by a strongly measurable, Bochner-integrable bounded generator produces an evolution family with the cocycle property; hence a nonzero CK deficit excludes a representation of that class. The manuscript expressly leaves outside this conclusion memory-kernel, Volterra, enlarged-state and more singular descriptions. The scoping is not cosmetic: the unqualified statement would be false, and the qualified one is exactly what the proof delivers. Section 3.2 separates the time-homogeneous regime (Hille–Yosida semigroup) from the genuinely non-autonomous regime (evolution family, Dyson series), a distinction on which several later statements depend.

3.3 Singular structure of the deficit (§4)

Remark 4.1. The methodological choice is correct and correctly justified: two-time transition operators are generically non-normal, compactness alone yields no eigenexpansion, and the Volterra operator on $L^2[0, 1]$ is compact with spectrum $\{0\}$ and empty point spectrum. The framework is accordingly built on Schmidt decompositions. Eigenstructure enters only through the covariant eigenfamilies of Definition 4.2, asserted where they exist, and Lemma 4.3 supplies the witness: under exact divisibility the multiplier of such an eigenfamily must be multiplicative, so nonmultiplicativity is a witness of indivisibility whenever an eigenfamily exists. This is logically sound, and the manuscript appropriately avoids claiming a general eigenexpansion. In my view the decision to build on singular rather than spectral structure is the single best methodological choice in the paper, and it is what makes the remainder of §4 possible.

§4.2, scope of compactness. The author gives a candid inventory of the classes where compactness holds (finite Ω; uniformly elliptic diffusions on bounded domains; L^2 kernels; nuclear

kernels) and where it fails (deterministic measure-preserving flows; point-mass kernels; translation kernels in infinite dimensions). The failures include several cases of considerable physical interest, and the paper says so rather than burying the point.

Theorem 4.5 and Corollary 4.6. The global subspaces

$$V_{\text{in}} = \overline{\text{span}} \bigcup_{s<r<t} \text{Range}\big(\Delta(t,s,r)^*\big), \qquad V_{\text{out}} = \overline{\text{span}} \bigcup_{s<r<t} \text{Range}\big(\Delta(t,s,r)\big),$$

are generated by the ranges of Δ^* and Δ over all triples. Standard Hilbert-space identities imply that each deficit operator vanishes on the orthogonal complement of the global input subspace, and for compact deficits the singular vectors supply extremal directions and a natural local geometry. The underlying range/kernel and singular-value facts are standard; their organization across the full CK-deficit family is the substantive step, and it is correct under the stated boundedness and compactness hypotheses.

The universal three-pair bound

$$D(t,s,r) \;\le\; \delta(t,s;\tilde{P}) + \delta(t,r;\tilde{P}) + \delta(r,s;\tilde{P}) \;\le\; 3 \sup_{(\tau_1,\tau_2)} \delta(\tau_1,\tau_2;\tilde{P}) \tag{2}$$

follows by adding and subtracting a divisible comparison family and applying the triangle inequality under the standing contraction hypothesis. It is operationally useful but mathematically elementary. Corollary 4.6 carries an explicit and correct scope limitation: the obstruction "concerns uniform operator-norm approximation of the two-time family itself; it does not preclude a divisible model matching selected observables, marginals, or low-order statistics of the same data." The constant is left open by the paper for optimization; it is settled below.

Theorem 4.8, the gauge-relative decomposition. Within a chosen unistochastic gauge, expansion of the squared modulus of a composed unitary separates diagonal path contributions from cross terms: $\Delta_{ij} = C_{ij} + D_{ij}$, with C the interference cross-term and D the squared-modulus non-composition gap. The identity holds for *every* gauge, and the critical point (which the manuscript makes explicitly) is that the two pieces are gauge-relative and only their sum equals the invariant CK deficit. Remark 4.9 supplies the witnesses: the same divisible family ($\Gamma \equiv F$ throughout) admits the gauge $(H, R(45°), R(45°))$ in which both terms are large with exact cancellation, and the gauge (KH, K, H) in which both vanish identically. Taken with the converse example of Remark 2.12, this prevents an illegitimate identification of one algebraic summand with an invariant physical interference quantity. The algebra itself is elementary; the value of the result is interpretive and organizational, and it is the pair of explicit gauge witnesses that gives it force.

Proposition 4.12, the indivisibility tensor. For a jointly C^2 family the manuscript defines $L(s) := \partial_t \Gamma(t,s)|_{t=s}$ and $C(s) := \partial_t^2 \Gamma(t,s)|_{t=s}$, and sets

$$\beta(s) \;:=\; C(s) - L(s)^2 - L'(s), \qquad L'(s) := \partial_s L(s), \tag{3}$$

proving that, uniformly in the position $\theta = (r-s)/(t-s) \in [0,1]$ and in s on the compact time simplex,

$$\Delta(t,s,r) \;=\; (t-r)(r-s)\,\beta(s) \;+\; o\big((t-s)^2\big), \qquad t-s \to 0. \tag{4}$$

Independent algebraic re-expansion confirms the coefficient. This is the paper's principal new object: a computable local invariant governing the exact leading-order scaling of the deficit, furnishing

a pointwise sufficient condition for indivisibility, and identifying through the singular structure of $\beta(s)$ the locally most deficit-active direction. The factorization $(t-r)(r-s)$ is the informative part of (4): it says the deficit vanishes to second order at both endpoints of the intermediate interval and is maximal at its midpoint, which is a structural statement and not a normalisation.

Finding 1 (What exactly is new in β, and what is not)**.** The calibration of this result requires care, and the paper does not currently supply it. For a genuine differentiable evolution family $U'(t,s) = A(t)U(t,s)$, the familiar identity $U'' = (A' + A^2)U$ gives $C = A' + A^2$ at the diagonal. Hence

$\beta \equiv 0$ is precisely the second-order compatibility condition expected of cocycle dynamics,

and the algebraic combination $A'+A^2$ (equivalently L^2+L') is not new: it is a standard consequence of differentiating a non-autonomous evolution equation twice, and is the second-order term of the Magnus/Dyson expansion (canonical reference: Blanes, Casas, Oteo & Ros, *Phys. Rep.* **470**, 151 (2009)). The paper states the identity without attribution.

What appears original, is the use of the *discrepancy* $C - L^2 - L'$ as a named CK-indivisibility tensor for arbitrary two-time stochastic transition families, together with the exact intermediate-time factor $(t-r)(r-s)$, the singular-direction interpretation, and the integration of the tensor into the global deficit theory. The revision should say exactly this: the expansion is textbook, the residual and its role are the contribution. Making the distinction explicit strengthens rather than weakens the originality claim, because it identifies precisely what is being claimed.

3.4 The Stinespring lift (§5)

The diagonal-retention channel is completely positive and trace preserving and reproduces the classical transition of populations; the Stinespring isometry is explicit, and the continuity argument uses standard dilation-continuity and projection/frame machinery. Under the strengthened square-root absolute-continuity hypothesis (R3′), the resulting unitary lift is absolutely continuous in time and admits an a.e.-defined integrable Hermitian logarithmic derivative. Theorem 5.4 accordingly delivers: (i) a strongly continuous lift with $W(s,s) = I$; (ii) a formal Hamiltonian generator under (R3′); (iii) an exact grid cocycle under divisibility; and (iv) minimal-dilation uniqueness, with the deleted set correctly taken to be vanishing *at the pair* (t,s), in general (t,s)-dependent and constant across $\{s < t\}$ only under (R3′).

Three remarks deserve notice. Remark 5.3 identifies and repairs a genuine under-determination: two unitaries reproducing the same *transition statistics* may dilate different *channels* — for $\Gamma = I$, take $W_1 = I$ against $W_2(|j\rangle \otimes |0\rangle) = |j\rangle \otimes |e_j\rangle$ — so the lift must be defined at channel level. Remark 5.6 grades each part of the theorem by level of rigour, which is unusual and commendable. Remark 5.8 confronts the divergence of $\widehat{H}_s(t)$ at the diagonal, where $\partial_t \sqrt{\Gamma_{ij}} \sim (t-s)^{-1/2}$ is locally integrable but unbounded, rather than concealing it.

Finding 2 (The "Hamiltonian recovered from the data" wording overstates the theorem)**.** The theorem is correct under its stated hypotheses; the issue is interpretive, and it should be fixed because the current phrasing invites a reading the mathematics does not support. The empirical kernel determines a *constructed* channel and permits construction of a unitary dilation, but neither the Stinespring representation nor the unitary completion is unique in any canonical physical sense (as the paper's own Remark 5.3 in effect concedes). What is established is the existence of a formal Hamiltonian associated with a *selected* continuous lift, not the recovery of a unique microscopic Hamiltonian. I recommend replacing "Hamiltonian recovered from the data" throughout

by wording such as "a formal Hamiltonian generator can be constructed for a selected continuous dilation". This is a matter of wording and interpretation, not of theorem validity.

Finite-grid cocycle dilation. The construction is formulated with the necessary distinction between the pairwise diagonal-retention channel $\Phi(t_m, t_l)$ and the composition of increment channels. Each grid increment receives a fresh environment, the global unitary is their ordered product, and the reduced increment-channel composition reproduces the divisible population transitions. This is a legitimate collision-model / process-tensor style construction, correct as a finite-grid statement, with the fixed-dimension continuum cocycle problem appropriately left open. Its priority status is discussed below.

3.5 Worked examples and causal structure (§§6–7)

Section 6 develops the two-state family $\alpha_\varepsilon(t,s) = e^{-\lambda(t-s)}\big(1-\varepsilon(t-s)^2\big)$ in closed form, and a three-state non-doubly-stochastic family exhibiting $V_{\text{in}} \neq V_{\text{out}}$, with an honest note that Assumption 3.3 fails for the latter.

Section 7 introduces the retarded transition operator on $L^2([0,T]\times\Omega)$, establishes causality, and proves Theorem 7.4: divisibility $\iff$ pointwise kernel-deficit vanishing $\iff$ $J(R) = 0$, with Lemma D.1 supplying the non-trivial passage from operator-level to pointwise vanishing. The triple integral of the squared operator norm of Δ is nonnegative and, under continuity, vanishes exactly when the deficit vanishes at every triple; the almost-everywhere to everywhere step is justified by continuity. The construction is a legitimate scalar consolidation of the two-time defect.

Finding 3 (The continuity hypothesis of Theorem 7.4 is correctly placed)**.** Hypothesis (C2) is imposed on the *open* region $\{s < t\}$, and Remark 7.5 gives the reason, which I have checked and regard as important. Operator-norm continuity on the *closed* simplex would be unsatisfiable in infinite dimensions under this framework's own standing assumptions: with $P(t,s)$ compact for $t > s$ and the boundary condition $P(s,s) = I$, the identity operator would be a norm limit of compact operators. A theorem carrying that hypothesis would be vacuous in precisely the setting the framework targets. Strong continuity is identified as the compatible notion at the diagonal, and I confirm that the proof uses continuity only on the open region. This is a hypothesis that is easy to state carelessly and consequential to state carelessly, and it is stated correctly.

3.6 The interpretive claim and its testability (§§2.8, 8)

Two statements bear on whether the framework meets the standard of logical consistency with a testable hypothesis, and both are, in my view, correctly handled.

Remark 2.8 states that unrestricted history augmentation always Markovizes a process, so that non-augmentability cannot be meant unconditionally; Bell-type arguments exclude only augmentations respecting locality and noncontextuality. This is the correct statement, and making it forecloses the strongest overclaim the framework invites.

Section 8 enumerates the competing mechanisms that a positive integrated deficit $J(R) > 0$ cannot separate — coarse-graining of a larger Markov system, omitted variables, nonstationarity, and measurement or estimation error among them — and adds a Davis–Kahan/Wedin caveat on the stability of estimated singular subspaces, noting that the subspace perturbation angle scales as the kernel estimation error divided by the relevant singular-value gap. The consequence should be stated plainly: the framework supplies a quantity that is estimable in principle but not, on the evidence of this paper alone, a discriminating test between indivisibility and ordinary hidden

structure. A nonzero deficit is moreover relative to the chosen state description. The author does not claim otherwise. The framework is internally consistent and $J(R)$ is a testable quantity; what is not yet supplied is a protocol under which a measured $J(R) > 0$ would discriminate between the candidate explanations, together with a null hypothesis against which it would be tested. Section 8 says as much and lists the estimator as open. This is a limitation honestly declared, not a defect concealed, and I do not treat it as an objection to publication — only as a boundary on what the paper establishes, and one that the empirical wording must continue to respect.

4 An open question resolved in the author's favour

Remark 4.7 poses as open whether the constant $1/3$ in Corollary 4.6 is optimal: "we do not claim it is optimal, and we know of no extremal divisible family attaining it. Determining the optimal constant over the class of divisible families is an open question." I have resolved it.

Finding 4 (The constant $1/3$ is optimal)**.** In the two-state symmetric reduction $\Gamma = \frac{1+a}{2} I + \frac{1-a}{2} X$ one has $\|\Gamma - \Gamma'\|_{\mathrm{op}} = |a - a'|$, and divisibility of a comparison family is the multiplicative law $\tilde{a}(t,s) = \tilde{a}(t,r)\tilde{a}(r,s)$. The best divisible approximation therefore solves

$$\min_{x,y} \ \max\big\{\, |a_{ts} - xy|,\ |a_{tr} - x|,\ |a_{rs} - y| \,\big\}.$$

Writing $x = a_{tr} + u$ and $y = a_{rs} + v$, the binding condition to first order is $D - (|a_{tr}| + |a_{rs}|)m = m$ with $m = |u| = |v|$, whence

$$\min_{\tilde{P}} \ \sup_{(\tau_1,\tau_2)} \delta(\tau_1, \tau_2; \tilde{P}) \ \approx \ \frac{D}{1 + |a_{tr}| + |a_{rs}|} \ \geq \ \frac{D}{3},$$

with equality requiring both intermediate kernels to be deterministic. An extremal family is explicit:

$$\Gamma(t,r) = \Gamma(r,s) = I, \qquad \Gamma(t,s) = I - \tfrac{D}{2}(I - X),$$

which is doubly stochastic with $\|\cdot\|_{\mathrm{op}} = 1$ — so Assumption 3.3 holds — and is indivisible with deficit exactly D. The computed ratio $\big(\text{true optimum}\big)/(D/3)$ takes the values 1.077, 1.036, 1.011, 1.0034, 1.0011 at $D = 0.6, 0.3, 0.1, 0.03, 0.01$ respectively, tending to 1; a search over 4,000 random triples never fell below 1.

The infimum of the ratio over admissible families is therefore 1, approached as $D \to 0$: the constant cannot be improved. On smooth families with non-deterministic intermediate kernels the bound is loose by a factor of roughly 1.1–2.5, which is its correct practical reading. The author should state both facts: the constant is sharp, and it is sharp only at a degenerate corner of the admissible class.

5 Originality: what is new and what is standard

Load-bearing and original.

- The indivisibility tensor $\beta(s)$ and Proposition 4.12, verified above in the general non-commuting, non-autonomous case — with the calibration of Finding 1: the underlying differential identity is standard, the named discrepancy, its triple-scaling law and its integration into the deficit theory are the contribution. This carries the greatest doctoral weight.

- The global subspace pair $(V_{\text{in}}, V_{\text{out}})$ as an intrinsically defined locus, with the three-state example demonstrating that the two genuinely differ. Defect spaces, ranges and singular subspaces are standard operator theory; I located no direct predecessor defining global input and output spaces by spanning the ranges of *all* CK deficits.
- The overall organization: a coherent defect geometry for CK-indivisible two-time families, which is where the strongest originality claim properly resides.

Moderate: interpretive or organizational novelty. The gauge-relative decomposition (Theorem 4.8 with Remarks 4.9 and 2.12) — the algebra is elementary and squared-modulus cross-term analyses are inherent in the stochastic–quantum correspondence and in unistochastic representations; what the paper adds is the demonstration, by explicit witnesses, that only the sum is invariant. Covariant eigenfamilies — moderate novelty in using multiplier nonmultiplicativity as a CK witness, but not a new non-autonomous spectral formalism. The integrated deficit $J(R)$ — the specific triple-integrated classical CK deficit appears distinct, but the general measuring principle is prior.

Correct, but short consequences of standard facts carrying more presentational weight than they support. Proposition 3.6 follows in two lines from the multiplicativity of time-ordered products; Theorem 4.5(a) is immediate from $\ker \Delta = \text{Range}(\Delta^*)^\perp$; Corollary 4.6 is a triangle inequality, albeit now shown to be optimal.

Substantially an assembly. Theorem 5.4 rests on Stinespring's theorem, the Kretschmann–Schlingemann–Werner continuity theorem, Kato parallel transport, and Steenrod/Michael continuous selection. The new ingredient is the *diagonal-retention indexing*, which routes all diagonal transitions through a single environment vector so that $\Phi(s, s) = \text{id}$ and a continuous unitary completion with $W(s, s) = I$ becomes possible. This is real and slightly delicate, but small relative to the theorem's presentation, and both continuum cocycle directions remain open.

6 Priority in the literature

The audit below distinguishes a *prior mathematical ingredient* from a *direct prior construction.* A result may be built from standard tools and still be original as a new theorem in a new object class; conversely, a technically sophisticated construction may have low priority if the same structure was already established in the relevant literature. The reference point for chronology is the deposit of 21 May 2026.

Element	Prior record	Assessment
Non-autonomous Koopman theory for two-time families	Maćešić, Crnjarić-Žic & Mezić, *SIAM J. Appl. Dyn. Syst.* **17**, 2478 (2018); Crnjarić-Žic, Maćešić & Mezić, *J. Nonlinear Sci.* **30**, 2007 (2020)	Cited, and correctly characterised as *presupposing* the cocycle property. Removing that hypothesis is a genuine extension. "Koopman spectral theory" in the title must accordingly be read broadly: the original contribution is the defect-based extension, not non-autonomous Koopman theory as such
Non-autonomous transfer operators	Froyland and collaborators (2010, 2013, 2023)	Cited, correctly
Approximate-flow and sewing theory	Rough-path and nonlinear sewing theory has a mature vocabulary for families that are only approximately multiplicative, with explicit estimates on $\mu_{ts} - \mu_{tu} \circ \mu_{us}$	**Not cited** (verified against the paper's reference list). The abstract idea of measuring failure of a cocycle law by a defect cannot be claimed as new. The purpose differs — sewing treats a small local defect as an error to be controlled so that an exact flow can be reconstructed, whereas here the defect is the object of study — and I located no prior formulation of the latter package. But the neighbouring literature should be acknowledged. [**R**]
Process-tensor and multi-time dilation; collision and repeated-interaction models	Pollock et al., *Phys. Rev. A* **97** (2018), 012127; Milz & Modi (2021); Costa & Shrapnel (2016); Ciccarello et al., *Phys. Rep.* **954**, 1 (2022)	Cited, with four structural distinctions stated, and the fresh-environment-per-step structure correctly identified as the standard collision model. The finite-grid cocycle dilation is technically valid but should not be presented as a flagship original theorem. [**R**] (scope of claim only)

Element	Prior record	Assessment
Divisibility-based non-Markovianity measures	Wolf, Eisert, Cubitt & Cirac (2008); Breuer, Laine & Piilo (2009); Rivas, Huelga & Plenio (2010, 2014); Chruściński & Maniscalco; Buscemi & Datta (2016)	Cited, with the distinction to $J(R)$ stated (column-stochastic directly; triples rather than pairs). Neither divisibility as a structural criterion nor scalar measures of divisibility failure are new; the contribution lies in the operator geometry of a *classical* two-time family
Unistochasticity; Birkhoff polytope	Bengtsson, Ericsson, Kuś, Tadej & Życzkowski (2005); Bengtsson & Życzkowski (2017)	Cited; standard facts used correctly
Stinespring continuity	Kretschmann, Schlingemann & Werner (2008); Shirokov (2020)	Cited; Shirokov correctly used to note that strong convergence of isometries does not lift to unitary completions. Competent application, not a priority claim
Embedding of classical stochastic matrices/kernels into CPTP maps	**J. Doukas**, "On the emergence of quantum mechanics from stochastic processes", arXiv:2602.22095 (25 Feb. 2026); **B. Canturk**, "Unification of stochastic matrices and quantum operations for N-level systems", arXiv:2503.09333 (2025), *Annals of Physics* **487** (2026), 170367	**Neither cited.** Both predate the deposit and both do exactly this. **[R]**, and the most consequential item in this audit — see Finding 5
$L(s)^2+L'(s)$ as the second-order coefficient of a time-ordered exponential	Standard second-order term of the **Magnus/Dyson expansion**; Blanes, Casas, Oteo & Ros, *Phys. Rep.* **470**, 151 (2009)	**Not cited**; stated without attribution (Finding 1). **[R]**
Contemporaneous neighbouring work	**J. Tian**, "Compression Covariance and Tangent kernels", arXiv:2606.12561 (10 June 2026): a compressed semigroup whose failure of the semigroup law is an operator-valued defect, treated as a kernel, with short-time rescaling yielding a tangent kernel	Postdates the deposit and therefore does *not* defeat priority. It does show that the mathematical neighbourhood is active, and that broad claims such as "first operator theory of semigroup defects" would be unsafe. **[R]** (framing only)

Finding 5 (The stochastic-to-quantum lift should not be claimed)**.** This is the area in which priority must be most carefully limited. Stinespring dilation is itself foundational. Beyond that, work predating the deposit already embeds classical stochastic kernels and matrices into CPTP maps and relates CK divisibility to quantum divisibility. Doukas (February 2026) lifts an arbitrary finite-dimensional stochastic kernel Γ to a map on $B(\mathcal{H})$, formulates the lift-compatibility relation, gives "an explicit dictionary between Γ and CPTP (Kraus) maps", and isolates Chapman–Kolmogorov divisibility of the lifted family as the decisive additional constraint. Canturk unifies the evolution of one-point probability vectors for stochastic and quantum processes for N-level systems and proves that CP-divisibility of quantum operations is the natural extension of the Chapman–Kolmogorov equation.

I have read both at source. Neither is cited. The manuscript should therefore not claim

priority for the basic stochastic-matrix-to-channel lift. Its defensible contribution in §5 is narrower and should be stated as such: the particular diagonal-retention implementation, the continuity/completion analysis for its two-time family, and the way the lift is incorporated into the deficit framework.

Finding 6 (Priority conclusion)**.** No prior work was located that reproduces the complete combination of (a) the CK composition deficit as the central operator on a two-time stochastic family, (b) global singular input/output deficit subspaces, and (c) the local tensor β with the stated triple-scaling law. The strongest originality claim is credible at the level of this combined framework. By contrast, Stinespring dilation, non-autonomous Koopman families, divisibility measures, approximate-flow defects, and finite-grid repeated-interaction dilations all have substantial prior art.

Priority rating: strong, but not absolute. The paper appears original in its principal synthesis and in specific defect diagnostics; several subsidiary constructions are applications or adaptations of established theory.

7 Specific criticisms

Items are marked **[M]** where they bear on the correctness or the substance of a claim, and **[R]** where they concern presentation, attribution or scope and are addressable by revision.

[R] Priority language.
Any use of "first", "new spectral theory" or equivalent broad priority language should be tied specifically to the defect-based construction, and not to non-autonomous Koopman theory, Stinespring dilation, or stochastic-to-CPTP lifting generally (Findings 5 and 6).

[R] Stochastic-to-quantum embeddings not cited.
Doukas (arXiv:2602.22095) and Canturk (arXiv:2503.09333; *Ann. Phys.* **487**, 170367) both predate the deposit and both construct the lift the paper presents. They should be cited and the claim narrowed.

[R] Magnus/Dyson attribution gap
for the expression $L(s)^2 + L'(s)$ (Finding 1).

[R] Approximate-flow / sewing literature not acknowledged.
It is the neighbouring defect-of-composition literature, and acknowledging it — while stating the difference of purpose — would strengthen rather than weaken the positioning of β and Δ.

[R] Formal Hamiltonian wording
(Finding 2). The current phrasing may suggest uniqueness or physical identifiability that is not proved.

[R] Proportion between claim and content.
Proposition 3.6, Theorem 4.5(a) and Corollary 4.6 are presented with more weight than their content carries. The paper already applies the correct remedy to Proposition 2.6 and Corollary 2.7 — explicit relabelling as scope-fixing background — and the same treatment should be extended.

[R] Theorem 5.4 is largely an assembly,
and its presentation does not distinguish the new ingredient (diagonal-retention indexing) from the imported machinery as sharply as Remark 5.6 distinguishes levels of rigour.

[R] Empirical interpretation.
Maintain, and do not weaken in revision, the caution that a nonzero deficit is relative to the chosen state description and does not by itself distinguish intrinsic memory from omitted variables, coarse-graining, nonstationarity, or estimation error.

[R] Compactness excludes
the deterministic and unitary cases, several of the physically most interesting. The paper declares this; it remains a limitation on the framework's reach.

[R] The examples are two- and three-state,
the §8 empirical protocol is a sketch, and the EDMD-type estimator that would operationalise it is listed as open. Nothing in the paper exercises the framework on a case where the answer is not evident in advance.

[R] Title and scope.
The title is defensible given the explanation supplied in §1, but "operator theory of CK-indivisible transition families" would be more exact. This is not a required correction.

[R] Remark 4.7 is superseded
by Finding 4.

I identified no material [M] defect. Every claim I verified is correct as stated. The criticisms above are matters of priority calibration, interpretive wording, literature positioning, and scope of demonstration; they do not invalidate any principal theorem.

8 Questions to the author

1. Explain precisely what is claimed to be original about $\beta(s) = C(s) - L(s)^2 - L'(s)$, given that $U'' = (A' + A^2)U$ is standard for differentiable non-autonomous evolution families, and attribute the expansion to Magnus/Dyson. State explicitly that the new content is the residual, the triple-scaling factor $(t - r)(r - s)$, and the integration into the deficit theory.

2. Confirm or refute Finding 4 — that the constant $1/3$ in Corollary 4.6 is optimal, with extremal family $\Gamma(t, r) = \Gamma(r, s) = I$, $\Gamma(t, s) = I - \frac{D}{2}(I - X)$, the infimum being approached as $D \to 0$ — and if confirmed, incorporate it, closing Remark 4.7. State also that the bound is attained only at a degenerate corner of the class, so that its practical looseness on smooth families is not misread.

3. State the strongest priority claim you would defend for V_{in} and V_{out}, and identify the closest operator-theoretic notion of a defect space that you consider prior art.

4. Address Doukas (arXiv:2602.22095) and Canturk (arXiv:2503.09333; *Ann. Phys.* **487**, 170367), both predating the deposit, and state what remains your own in §5 once the basic stochastic-matrix-to-CPTP lift is conceded to that literature.

5. Clarify whether any feature of the formal Hamiltonian is invariant under changes of Stinespring dilation or of unitary completion. If not, which quantities derived from the lift are intended to carry physical status?

6. Explain the exact relationship between your finite-grid cocycle dilation and standard process-tensor or repeated-interaction/collision-model dilations.

7. Give a concrete statistical protocol by which Δ or β could be estimated from finite transition data, and state what null hypothesis would be tested. In doing so, address the conditioning problem recorded in the verification record above: β is a small difference of large second-order quantities, so any estimator will lose several significant digits to cancellation. State what accuracy in Γ would be required to resolve β at a given size.

8. Describe what an empirically measured CK violation can and cannot establish about intrinsic memory once hidden-state augmentation and coarse-graining are allowed.

9. Exhibit the framework on at least one family of dimension greater than three, or state why the two- and three-state cases are structurally sufficient.

9 Technical conclusion

The mathematics is correct so far as I have been able to check it, the central invariant is genuinely new in its formulation and diagnostic role, and the hypotheses are stated at the level the proofs require. The manuscript demonstrates real command (rather than terminological familiarity) across two-time stochastic kernels, conditional-probability structure, Hilbert-space operator adjoints, compact-operator singular theory, perturbation estimates, non-autonomous evolution families, second-order Taylor expansions, unistochastic matrices, Kraus/Stinespring representations, continuity of projection families, and causal integral operators. It also displays sound research discipline in separating classical background from claimed contributions, distinguishing pairwise channels from their grid compositions, restricting the no-generator claim to the class actually proved, and explicitly identifying open continuum-dilation questions.

The revisions called for are matters of priority calibration, attribution, interpretive wording and scope of demonstration, together with the incorporation of Finding 4. None requires new mathematics, and none disturbs the paper's principal results.